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The role of learners' socioeconomic status and perception of technology use in their second language learning motivation and achievement

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Abstract

As digital technology is becoming an increasingly critical component of education, learners' socioeconomic status (SES) and technology use are becoming more important in their learning. In the context of English learning, the present study modeled how learners' SES and their perception of technology use for language learning are linked to their motivation for learning English as a foreign language (EFL) and level of English achievement. To this end, 306 South Korean college students' learner factors, motivation, and English achievement were measured. The results of the structural equation modeling analysis indicated that learners' SES was positively associated with their perception of technology use and motivation and indirectly associated with their language learning achievement. Furthermore, we found that learners' perception of technology use in language learning is associated with their motivation and achievement. In addition, among the several learner factors measured in this study, learners' expectancy of being successful in language learning as part of achievement motivation emerged as the strongest predictor of their language learning achievement. An in-depth discussion of pedagogical and policy implications is also provided.

Keywords: Socioeconomic status (SES); motivation; computer-assisted language learning (CALL); English as a foreign language (EFL); structural equation modeling (SEM)

Introduction

The value of digital technology as an effective tool for language teaching and learning has been confirmed in critical and retrospective reviews (e.g. Chapelle, 2001; Chun et al., 2016; Salaberry, 2001; Warschauer, 2000) as well as in meta-analysis studies (e.g. Golonka et al.,

2014; Sung et al., 2015; Zhao, 2004). In addition, as Chun et al. (2016) noted, digital technology has become the norm in everyday life, and language learning environments without technologies are difficult to find today. Moreover, the ubiquity of technology use for language learning even extends its value to outside the classroom, where it enables learners to better learn and practice languages (Chapelle, 2003; Terhune, 2016), a quality that is all the more relevant in the era of the COVID-19 pandemic. Nevertheless, more research is needed to understand the role of the learner's background considering the persistent digital divide (i.e. the technology gap between rich and poor; for a recent meta-analysis on this topic, see Scherer & Siddiq, 2019). As part of this effort, a handful of recent studies have investigated ways to address the gap by seeking to understand the role of language learners' background factors, such as motivation (e.g. Lai, 2013; Lee et al., 2017) and socioeconomic status (e.g. Ghorbani & Golparvar, 2020).

To the best of our knowledge, however, modeling the relationship among these learner factors has not been attempted, and this is what we aim to achieve in the present study. Specifically, we formulate a structural equation model under the framework of the Expectancy–Value Theory (EVT) of motivation suggested by Eccles and colleagues (Eccles et al., 1983; Eccles & Wigfield, 2002; Wigfield & Eccles, 2000), in which learners' achievement motivation is assumed to comprise two components: (1) learners' expectation of success in a target task and (2) the value they attach to this success. This framework has been suggested to have considerable explanatory value in accounting for learners' motivation and learning outcomes (for a review, see Barron & Hulleman, 2015), and it is for this reason that we adopted this framework in the present study. In so doing, we focus on the role of learners' socioeconomic status (SES), as it has been suggested to be associated with their perception of technology use (e.g. Tondeur et al., 2011; Vekiri, 2010) owing to the persistent digital divide. This attempt may contribute to a deeper understanding of the relationship among language learners' SES, perception of technology use in learning the target language, language learning motivation, and language learning achievement.

Literature review

Technology use in language learning

Most research on the use of technologies in language learning (e.g. Benson, 2011; Ghorbani & Golparvar, 2020; Lai, 2013, 2015, 2019; Lee et al., 2017) has drawn on self-directed learning as an overarching framework. According to Knowles (1989), self-directed learning can be operationalized as 'a process in which individuals take the initiative … in diagnosing their learning needs, formulating goals, identifying human and material resources, choosing and

implementing appropriate learning strategies and evaluating learning outcomes' (p. 18), implying some level of learner control (Ozer & Yukselir, 2021). Because using technologies for language learning implies using resources that encourage learners to learn target languages on their own (Benson, 2011), the increasing support for self-directed learning, which has involved a shift from teacher-centered to student-oriented learning (Candy, 1991), has coincided with the increasing implementation of technology use for language learning inside and outside the classroom (Lai, 2013).

Because the learner is considered central to technology-mediated self-directed language learning, considerable research has focused on learners' perception of technology use and on contextual factors as predictors for learners' engagement with technologies for language learning. For example, previous language learning literature (e.g. Lai, 2013, 2015; Lai et al., 2012; Lai & Gu, 2011) has suggested several important learner factors, such as facilitating conditions (i.e. learners' perceived availability of related resources and support to facilitate their computer use for successful language learning), computer self-efficacy (i.e. learners' belief in their capacity to use computers for successful language learning), and perceived usefulness of computers (i.e. learners' perception of the usefulness of computer use for successful language learning), which are believed to determine learners' computer use in language learning.

In one of the earliest studies on this issue, Lai et al. (2012) examined relationships between undergraduate Hong Kong English-learners' computer use for their language learning and related learner factors (e.g. perceived usefulness of computers, computer self-efficacy, facilitating conditions, attitude toward computer use, and educational compatibility). The results revealed that facilitating conditions and participants' perceived usefulness of computer use successfully predicted their adoption of computer resources. Based on this finding, Lai (2013) developed a model of self-directed computer-assisted language learning (CALL), meant to reveal a range of attitudinal and sociocultural predictors for Hong Kong undergraduate learners' engagement with computers for language learning. In analyzing the survey data, Lai found that learners' perceived usefulness of technologies, language learning motivation, and educational compatibility were the dominant learner factors affecting their use of computers for out-of-class English learning. More recently, via interviews and surveys with undergraduate language learners, Lai (2015) found that three learner factors—facilitating conditions, computer self-efficacy, and perceived usefulness of computers—determined learners' self-directed use of computers for language learning.

Socioeconomic background and language learning

In the present study, we hypothesized that learners' perception of computer use for successful language learning would mediate the relationship between language learning motivation and their SES, which we discuss next. Despite the fact that digital technology is pervasive in everyday life (Chun et al., 2016), studies have emphasized the continued existence of a digital divide, that is, the idea that one's access to technologies may be influenced by one's SES background (e.g. Cho et al., 2003; Scherer & Siddiq, 2019), which is a construct for 'individual or group social prestige' (Ghorbani & Golparvar, 2020, p. 608). Although different ways of operationalizing and measuring SES have been proposed in the literature on the effects of SES on young people's lives and development (e.g. Entwislea & Astone, 1994; Hauser, 1994), SES has been frequently measured by family income and/or parental education in the education literature (e.g. Butler, 2014; Fernald et al., 2013; Ghorbani & Golparvar, 2020); we therefore operationalized this construct by considering both of these variables (see *Measurements* section for details).

Although this construct has received less attention in the field of language learning, some studies have been recently conducted to examine the effects of learners' SES on their language learning, which we review below. A study by Butler and Le (2018) collected data in three different phases (i.e. three years in total) from Chinese middle school students and their parents. The results showed that the higher SES parents maintained their expectations about their children's English learning ability, while the expectations of the lower SES parents decreased over the period of the study. The results also showed that 'both parental income and educational level were positively related to students' English performance' (p. 12), pointing to the important role of learners' SES background in learning another language. In a study by Huang et al. (2018) of Taiwanese high school students and their parents, the students were asked to perform a story-telling task, and both the students and parents completed a survey on the students' previous learning experience (i.e. type and amount of English input they had received). The results showed that SES was positively correlated with the amount of exposure to English in their pre-school years and with the experience of traveling abroad. SES also predicted English speech production outcomes but only in a rather limited way in the 'accuracy' dimension, through the mediation of an input-related variable. In their suggestions for future research, Huang et al. (2018) indicated the possibility that other learner-related factors, such as 'learners' motivation and self-efficacy', may mediate the relationship between students' SES and language learning outcomes (p. 34).

Kormos and Kiddle (2013) examined differences among secondary students in Chile from five different social classes in terms of variables related to learner motivation. The results showed that social class had a larger effect on learners' self-efficacy than other factors such as

motivation, self-regulation, parental encouragement, and independent use of technology, which were also found to be different among social classes.

More recently, Ghorbani and Golparvar (2020) aimed to model the relationship between SES, level of self-initiated use of computers, and language learning achievement among Iranian university students of English. Their model suggested that SES was a significant predictor of self-initiated use of computers, which was, in turn, positively associated with the participants' performance on the English section of their university entrance exam. Although this study deepened our understanding of the relationships among language learners' SES, adoption of computer resources, and language learning outcomes, it did not explore how learners' motivation mediates these relationships. Overall, however, SES appears to be an important learner factor that deserves more attention in language learning research, particularly in relation to learners' access to technologies and their perception of technology use.

Motivation to learn English as a foreign language

According to the EVT framework (Eccles & Wigfield, 1995, 2002; Wigfield & Eccles, 2000), the connection between learner factors and learners' achievement is formed by their achievement motivation. Although this framework has not been widely used in the field of language learning, as Barron and Hulleman (2015) suggested in their review, the EVT framework is highly rated in terms of its ability to integrate different theoretical perspectives on motivation, identify teachable and modifiable motivational factors, and explain learners' achievement-related behaviors, such as their choices, persistence, and performance. That is, it suggests that language learners' expectations of success in language learning (i.e. *expectancy*) and the amount of value they attach to language learning (i.e. *value*) determine their motivation, which, in turn, influences their language learning achievement as well as related choices, including their interest and effort devoted to language learning.

It has been suggested that these two major elements of the EVT framework, expectancy and value, are 'differentiated constructs' that 'independently influence individuals' choice of activities and individuals' behavior in various activity settings' (Eccles et al., 1993, p. 839). Wigfield and Eccles (2000) suggested that expectancy, in theory, could be divided into ability beliefs, which 'are defined as the individual's perception of his or her current competence at a given activity' (p. 70), and expectancy for success, which is more related to future events. While these two constructs are theoretically distinguishable, the findings of previous studies (e.g. Eccles & Wigfield, 1995) suggest that they may load onto one factor in empirical data.

It has been suggested that *value* comprises several components, including attainment value, intrinsic value, and utility value. Attainment value refers to the degree of importance one attaches to the task at hand (Eccles et al., 1983), while, as Wigfield and Eccles (2000)

acknowledged in their account of these components, the intrinsic and utility values are theoretically similar to intrinsic and extrinsic motivation in self-determination theory (Deci & Ryan, 1995), with intrinsic value resulting from enjoyment of the target task itself and utility value being related to doing the task for external reasons.

Thus, it has been suggested that expectancy and value each predict different aspects of learning better (e.g. performance, intention to continue learning; see Schunk et al., 2008; Wigfield & Eccles, 2000), with the expectancy component relating more to performance and achievement and the value component predicting one's persistence in studying the target subject.

Based on the above, we can assume that learners with high expectancy and value are more likely to put more effort to their language learning, be more deeply engaged, persist longer, use more appropriate strategies, and achieve higher learning outcomes (Eccles & Wigfield, 1995, 2002; Rueda & Chen, 2005). In addition, a major implication of research based on EVT lies in its domain-specific nature (Eccles et al., 1993), meaning that learners' expectations of success and achievement values may differ across different school subjects or activities (e.g. math, reading, sports). For this reason, EVT may be preferred over other motivational frameworks for domain-specific language learning tasks and for capturing learners' perception of their learning in sociocultural contexts (e.g. Rueda & Chen, 2005; Wen, 1997).

Present study

As part of an effort to understand the role of learners' SES and their perception of technology use in the CALL context, we attempted to model the relationships among these important learner factors using the EVT framework of motivation, as shown in Figure 1. First, we focused on learners' SES background, to check whether our participants' different backgrounds affected their perception of technology use for language learning, their motivation, and, ultimately, their language learning achievement. Further, using the EVT framework, we explored how learners' SES backgrounds and perception of technology use were correlated with their motivational factors, including their expectancy of being successful in learning a target language and their perceived subjective value of engaging in language learning tasks, which both contribute to their language learning achievement. The following is the primary research question that guided the present study.

RQ. What are the structural relations among learners' SES background, perception of technology use, motivation, and achievement in the context of CALL?

Based on the previous literature, our hypotheses for the research question were as follows:

1. The two motivational factors (i.e. expectancy and value) would directly predict language learning achievement.
2. The learners' perception of computer use for successful language learning would directly predict two motivational factors (i.e. expectancy and value); also, it would indirectly predict language learning achievement via these motivational factors.
3. SES would directly predict the learners' perception of computer use for language learning as well as two motivational factors (i.e. expectancy and value), and this relationship would mediate the indirect contribution of SES to language learning achievement.

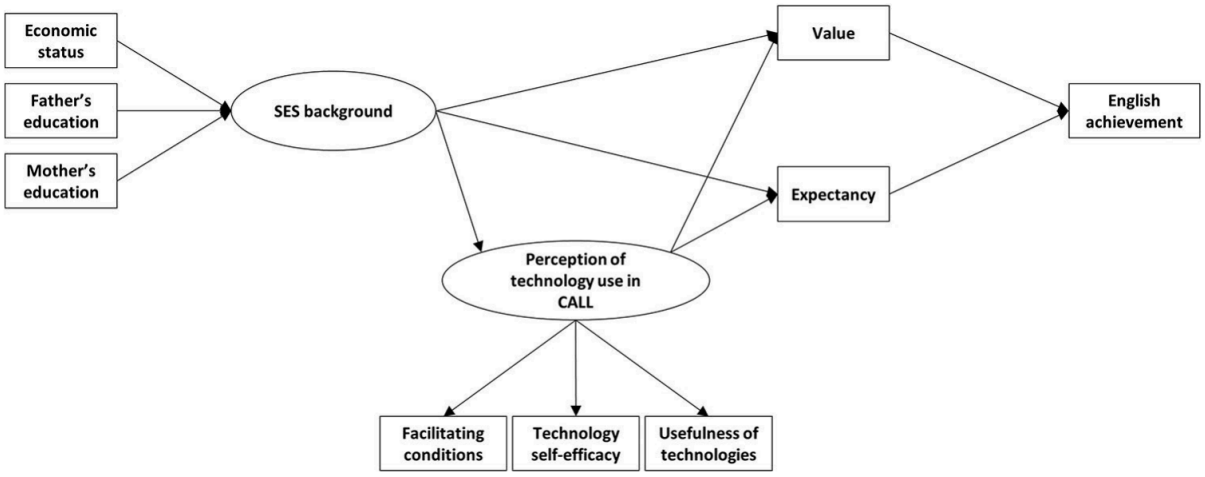


Figure 1. Hypothesized structural equation model.

Methods

Participants

A total of 306 South Korean college students registered in a one-semester English conversation course, geared toward intermediate-level learners, participated in this study. They were majoring in different academic disciplines: humanities (n = 67), social sciences (n = 70), engineering (n = 78), and sciences (n = 91). On average, they had approximately 12 years of experience in learning English in formal school contexts (four years in elementary school, six years in junior high and high school, and two years in college).

At the beginning of the semester, the participants filled in an informed consent form and received a short introduction to the data collection process. They were told that they could withdraw their participation in the study at any time without any consequences. They participated in surveys conducted on their SES background (economic status, parents' educational backgrounds), perception of technology use in CALL (facilitating conditions, technology self-efficacy, perceived usefulness of technologies), and motivational factors

(expectancy, value). All the participants completed the surveys (i.e. there were no missing values) and were financially compensated for their time. After completing the course for the semester, which included using technology resources inside and outside the classroom for their language learning, the participants took the Test of English for International Communication (TOEIC) developed by the Educational Testing Service to measure their language learning achievement. According to their average score obtained in the TOEIC (M = 812.8, SD = 86.35), they were independent users of English (CEFR B2; Educational Testing Service, 2016).

Measurements

First, to obtain the participants' SES information, we had to choose a limited number of variables among previously suggested ones for SES, such as parents' occupation, parents' marital status, cultural traditions, free/reduced price school meals, etc. Given the feature of the Korean school context (e.g. free lunches for every student) as well as the practical constraint of collecting detailed information about the participants' parents, the participants were asked to respond to questions regarding their economic status and parents' educational backgrounds, based on a common practice at the Bureau of the Population and Housing Census, Statistics Korea (KoStat, 2020). We categorized the economic status variable into three values of (1) low, (2) medium, and (3) high, based on the standardized values for the combination of total family income and property bands. For parents' educational backgrounds, a total of four values were formed for two variables (mother and father): (1) no high school diploma, (2) high school diploma, (3) community college degree, and (4) bachelor's degree or above.

To determine learners' perception of technology resources, we followed Lai's (2013, 2015) findings on factors affecting learners' self-directed CALL, thus adopting the following three factors: (1) facilitating conditions, which refers to the perceived availability of support facilitating the use of technologies for language learning; (2) technology self-efficacy, which refers to the individual's perceived ability to use technologies for language learning; and (3) usefulness of technologies, which refers to the perceived usefulness of technology use for language learning. To measure these factors, we adapted Lai and colleagues' questionnaire items based on a 5-point Likert-scale (Lai et al., 2012; Lai & Gu, 2011), with some minor changes in the wording of the original items to enhance their content validity. Our collected data showed acceptable internal reliability based on a total of 10 items: three items for facilitating conditions (α = .83), three items for technology self-efficacy (α = .85), and four items for perceived usefulness (α = .85) (see the Appendix for the survey items).

Regarding the two motivational factors (i.e. expectancy and value), we adapted Eccles and Wigfield (1995) questionnaire items, whose internal reliability has been reported to be between .70 and .86 across different subjects, such as math, reading, music, and sports (Eccles et al., 1993). The original 5-point Likert-scale items in Eccles and Wigfield's study were contextualized to English learning for the sake of content validity. The expectancy component was measured through five items on learners' expectancy of being successful in English learning ($\alpha = .89$) and the value component through seven items on learners' perceived value of engaging in English learning tasks ($\alpha = .83$) (see the Appendix for the survey items).

To assess the participants' English learning achievement after taking the one-semester-long English course, they took the TOEIC, which consists of two sections (listening and reading comprehension). The TOEIC was considered to have adequate validity as a measure of the participants' English learning achievement, as obtaining a particular TOEIC score or higher was a prerequisite for the sampled students' graduation; therefore, this score may be highly related to the two motivation components (Wigfield & Eccles, 2000). This test takes approximately 2 hours: 45 minutes for the listening section and 75 minutes for the reading section. Each section is scored from 5 to 495 points, making the maximum score 990. Detailed information on the types of questions in the listening and reading sections can be found on the Educational Testing Service website (https://www.ets.org/toeic/organizations/listening-reading/about/content-format).

Data analysis plan

Structural equation modeling (SEM) using STATA 14 software (StataCorp, 2015) was the primary analytical method in the present study. In the proposed model, to measure the latent variable for learners' SES background, we included the participants' economic status and father's and mother's education backgrounds. Although these three variables form a construct for SES, they were not highly correlated ($r < .7$); consequently, we made this construct a composite latent variable (or a formative construct), whose error variance was therefore fixed at zero (Acock, 2013). To construct the latent variable for learners' perception of technology use, we included the following three factors: facilitating conditions, technology self-efficacy, and perceived usefulness. Lastly, learners' expectancy of being successful in language learning and value of engaging in language learning practices were included as the two motivational factors, and the TOEIC score variable was included as the learners' English achievement in the model. To ensure the goodness of fit of the SEM model, we adopted five model fit indices suggested by Kline (2012) and previous literature (Acock, 2013; Lee et al., 2020): the chi-

squared test (a testing model should not significantly differ from a saturated model, $p > .05$), root mean square error of approximation (RMSEA < .05), comparative fit index (CFI > .95), Tucker–Lewis index (TLI > .95), and standardized root mean square residual (SRMR < .05).

Results

Descriptive statistics

Table 1 presents the descriptive statistics for the observed categorical variables, such as the categorical values, sample sizes, and proportions, and Table 2 includes the descriptive statistics for the observed continuous variables, such as the means, standard deviations, and internal reliabilities. As shown in Table 1, we found that the majority of the participants considered the economic status of their families to be middle class (n = 283, 92%), while only 3% (n = 8) and 5% (n = 15) reported the economic status of their families as low or high income, respectively. Regarding their parents' educational backgrounds, the majority of the participants (n = 237, 77%) reported their fathers to have either community college degrees (n = 16, 5%) or bachelor's degrees or above (n = 221, 72%), while about 23% reported that their fathers did not have college degrees (n = 69). As for their mothers, about 62% had college degrees (n = 191). According to the correlation analysis, there was a moderate correlation between fathers' and mothers' educational backgrounds (r = .57, $p < .001$) and weak correlations between economic status and fathers' and mothers' educational backgrounds (r = .20, $p < .001$; r = .15, $p < .001$, respectively). For this reason, when forming a latent variable for learners' SES background, a composite latent variable (i.e. a formative construct) was formed.

Table 1. Descriptive statistics of the observed categorical variables.

Variable	Category	Sample size	Proportion
Economic status	Low	8	3%
	Middle	283	92%
	High	15	5%
Father's education	No high-school diploma	5	2%
	High-school diploma	64	21%
	Community college degree	16	5%
	Bachelor's degree or above	221	72%
Mother's education	No high-school diploma	8	3%
	High-school diploma	107	35%
	Community college degree	27	9%
	Bachelor's degree or above	164	53%

Table 2. Descriptive statistics of the observed continuous variables.

Variable	Mean (SD)	α (# of items)	Normality test[a]
Motivational factors			
Value	3.84 (.65)	.83 (7)	Passed at 1%
Expectancy	2.95 (.89)	.89 (5)	Passed at 1%
Perception of technology use			
Facilitation conditions	3.28 (.90)	.83 (3)	Passed at 5%
Technology self-efficacy	3.26 (.90)	.85 (3)	Passed at 5%
Usefulness of technologies	3.76 (.72)	.85 (4)	Failed[b]
English achievement			
TOEIC	812.80 (86.35)	n/a	Passed at 5%

[a]We used the Jarque and Bera (1987) test for normality.

[b]Most of the participants found the technology very useful in their L2 learning. To resolve any statistical bias coming from its non-normality, we used a robust estimator (also known as a distribution-free linearized estimator) for the structural equation modeling analysis.

In Table 2, we constructed the value (based on seven questionnaire items with an internal reliability of .83) and expectancy (based on five questionnaire items with an internal reliability of .89) variables for motivation; their means and standard deviations were 3.84 (*SD* = .65) and 2.95 (*SD* = .89), respectively. For learners' perception of technology use, we constructed a latent variable based on a total of three variables: facilitating conditions (based on three questionnaire items with an internal reliability of .83), technology self-efficacy (based on three questionnaire items with an internal reliability of .85), and perceived usefulness of technology use (based on four questionnaire items with an internal reliability of .85); their means (and standard deviations) were 3.28 (*SD* = .90), 3.26 (*SD* = .90), and 3.76 (*SD* = .72), respectively. For English achievement, the participants' TOEIC scores were calculated, and the mean (and standard deviation) was 812.80 (*SD* = 86.35).

To ensure robustness of our SEM model, we checked whether the dependent variables in the proposed model were normally distributed. According to the results of the Jarque and Bera (1987) test for normality, as shown in Table 2, all dependent variables except for usefulness of technologies were normally distributed. To resolve any possible statistical bias coming from the non-normality of the usefulness of technologies variable, where most participants reported that they found technology very useful for their language learning, we used a robust estimator (i.e. a distribution-free estimator) for the SEM analysis, according to West et al. (1995) suggestion.

Structural equation modeling

Prior to the SEM analysis, we checked whether our measurement model for learners' perception of technology use (that is, a reflective latent variable in the SEM model) had been validly measured by the three variables. The results indicated that the model had acceptable model fit indices, χ^2 (3) = 2.48 (p > .05), RMSEA < .05, CFI > .95, TLI > .95, and SRMR < .05, and that the standardized factor loadings were statistically significant (p < .001) and large enough, ranging from .60 to .83, to prove statistical power (Kline, 2012).

Furthermore, the results for the finalized model (see Figure 2), including the reflective latent variable for learners' perception of technology use, a composite latent variable for learners' SES background, two observed variables for motivational factors, and an observed variable for English achievement, indicated that our data had acceptable model fit indices, χ^2 (17) = 22.94 (p > .05), RMSEA < .05, CFI > .95, TLI > .95, SRMR < .05.

Figure 2 illustrates the complex associations among variables to model structural relations among learners' SES background, perception of technology use, motivation, and achievement in the context of CALL. The values are standardized path coefficients for the associations of variables; the solid lines represent statistically significant associations (p < .05), and the dashed lines represent non-significant associations (p > .05). Above all, the results indicated that the participants' economic status (β = .53, p < .01) and father's educational background (β = .74, p < .001) positively contributed to constructing the latent variable for learners' SES background. As noted, the latent variable was constructed as a composite variable because its three indicators—economic status, father's education, and mother's education—were not strongly correlated (r < .7); therefore, the three indicators have their own path coefficients. Overall, we found that the latent variable for learners' SES background had a positive association with the latent variable for learners' perception of technology use (β = .13, p < .05).

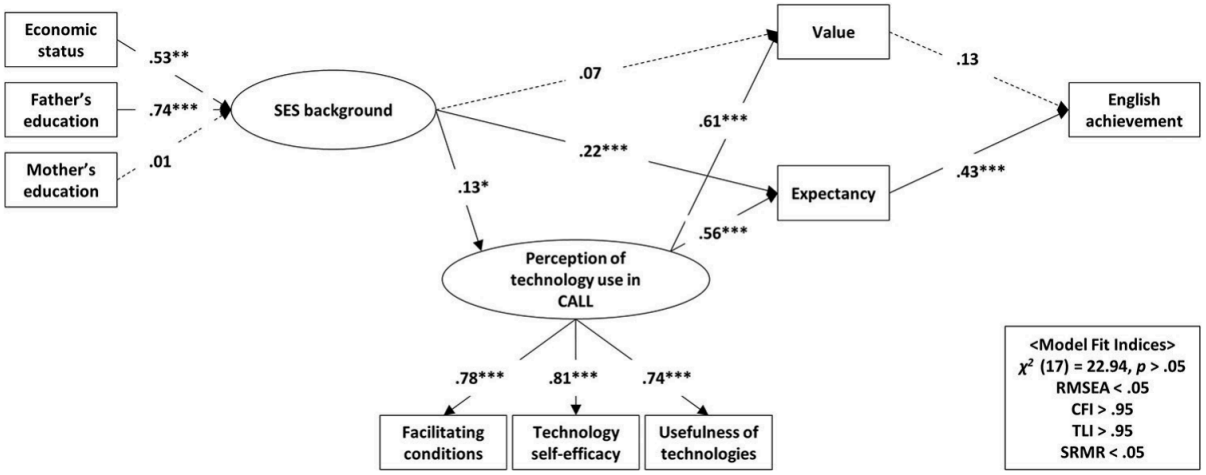


Figure 2. Finalized structural equation model.
Note. *p < .05, **p < .01, ***p < .001.

The latent variable for learners' perception of technology use was based on the three observed variables, facilitating conditions, technology self-efficacy, and perceived usefulness of technology use for language learning, with significant, strong factor loadings (β > .70, p < .001). Unlike the latent variable for SES background, the latent variable for perception of technology use was constructed as a reflective latent variable; in other words, its three indicator variables were used to construct this latent variable based on the computed factor loadings, and therefore, these indicators were not included individually in the path coefficient calculations (Acock, 2013).

The results revealed an association between the proposed learner factors (i.e. learners' SES background, perception of technology use) and the motivational factors. We found that the learners' SES background had a significant association with the expectancy component of motivation (β = .22, p < .001) but not with the value component. However, learners' perception of technology use had significant relationships with both value (β = .61, p < .001) and expectancy (β = .56, p < .001).

Lastly, regarding the relationship between motivational factors and achievement, of the two motivational factors, only expectancy had a significant association with the participants' English achievement (β = .43, p < .001).

The calculated standardized path coefficients shown in Figure 2 only present direct effects of the independent variables on the dependent variables, with indirect effects of each independent variable mediated by other variables on the dependent variables not being shown in Figure 2. Therefore, Table 3 presents the total effects, after including both the direct and indirect effects, of the independent variables on the dependent variables to further understand the complex relationships among the variables. As shown in Table 3, we found that the participants'

economic status and father's educational background had significant total effects on English achievement (β = .08, p < .01; β = .11, p < .01, respectively). In addition, the latent variable for learners' perception of technology use was found to demonstrate a significant total effect on the English achievement variable (β = .32, p < .001).

Table 3. Total effects of the independent variables in the SEM model.

Independent variables		**Dependent variables**			
		Perception of technology use	**Motivational factors**		**English achievement**
			Value	**Expectancy**	
SES background	ECON status	.07 (.04)	.08 (.04)	.15** (.05)	.08** (.03)
	Father's EDU	.10* (.05)	.11 (.06)	.22*** (.05)	.11** (.03)
	Mother's EDU	.00 (.03)	.00 (.04)	.01 (.08)	.00 (.04)
Perception of technology use	N/A[a]	–	.61*** (.06)	.56*** (.05)	.32*** (.04)
Motivational factors	Value	–	–	–	.13 (.07)
	Expectancy	–	–	–	.43*** (.06)

*p < .05, **p < .01, ***p < .001.

[a]As this is a reflective latent variable, its three indicators—facilitating conditions, technology self-efficacy, and usefulness of technologies—are used to construct the latent variable and are thus not included for path coefficient calculations.

Taking these results together, we found that four variables—economic status, father's educational background, latent variable for learners' perception of technology use, and expectancy—had statistically significant total effects on learners' English achievement. Further, for all independent variables contributing to English achievement, we conducted an additional data analysis, a likelihood-ratio test between coefficients in an SEM model, to measure their relative contributions to English achievement. The results showed that the expectancy variable (β = .43, p < .001) had the largest total effect on English achievement, followed by the latent variable for learners' perception of technology use (β = .32, p < .001), also with a significant difference (p < .05). Their contributions were significantly larger (p < .05) than those of economic status (β = .08, p < .01) and father's educational background (β = .11, p < .01).

Discussion

This study examined the relationship among Korean college students' SES background, perception of technology use, language learning motivation, and learning achievement in the context of computer-assisted English learning. The main findings are as follows:

1. SES directly predicted the learners' perception of technology use for language learning and their expectancy and indirectly predicted language learning achievement via expectancy.
2. Learners' perception of technology use for successful language learning directly predicted two motivational factors (i.e. expectancy and value), while it indirectly predicted language learning achievement.
3. Out of the two motivational factors, only expectancy could significantly predict language learning achievement.

Based on these findings, we believe that it is worth discussing (1) the role of learner factors in the context of CALL and (2) how the EVT framework could account for the motivation among our participants to learn the target language. Further, we will provide the pedagogical implications of our findings, followed by the limitations of the present study and suggestions for future research.

Role of learner factors in language learning

Our structural equation model indicated that learners' SES background had a direct effect on their expectancy of being successful, indicating that the higher SES participants had a stronger belief in their ability to learn English. This finding mirrors that of Kormos and Kiddle (2013), who found that social class greatly influenced English language learners' self-efficacy in Chile. As self-efficacy in their study is conceptually similar to the expectancy component in the EVT framework adopted in the present study, our finding supports Kormos and Kiddle (2013) proposition that 'socio-economic context plays an important role … in influencing students' self-efficacy beliefs and the effort and persistence with which they carry out learning tasks' (p. 402). Although our findings do not provide a concrete explanation of how learners' SES background influences their expectancy, a possible explanation may be derived from Butler and Le (2018), who found that higher SES parents continue to have high expectations from their children throughout their education, which may have positively contributed to these children's expectancy that they would be successful in their language learning.

Furthermore, our model indicated that learners' perception of technology use, which was influenced by learners' SES, is another significant predictor for both components of the motivational factors (i.e. expectancy and value), corresponding with Lai's (2013) finding that motivational factors have a significant link with perception and the use of technology resources among undergraduate Hong Kong learners. This finding can be interpreted as follows: having facilitating conditions for technology use and a high level of belief in one's ability to use technologies as well as in the perceived usefulness of technology use for language learning, which are influenced by SES, will not only positively influence the

development of learner autonomy and expectancy to achieve success in language learning (see Benson, 2011; Lai, 2015 for a similar discussion), but also help students to have a clearer image of the importance of learning English and thus attach more value to engaging in such a language learning task.

Learners' SES background was found to play a significant role as a learner factor in the context of CALL. This finding is in accordance with that of Ghorbani and Golparvar (2020), among Iranian EFL learners. Therefore, the findings of the present study and Ghorbani and Golparvar together suggest that learners with a higher level of SES may 'have access to better and more effective educational resources, including technological ones' for language learning (p. 619) and may accordingly have a more positive perception of them. The relationship between SES and the perception of technology use revealed in this study is also in accordance with the persistent digital divide, supported by the overall positive relationship between learners' SES and their ICT literacy found in a recent large-scale meta-analysis (Scherer & Siddiq, 2019).

Empirical evidence of the EVT framework in language learning

In our structural equation model, we found that two motivational factors in the EVT framework significantly mediated the relationships between learner factors and language learning achievement. This finding fits with the general predictions under the EVT framework (Eccles & Wigfield, 1995; Wigfield & Eccles, 2000), lending weight to the proposition, pertaining to this framework, that two motivational components are immediate precursors of achievement-related choices and learning outcomes with the latter, in this case, being our participants' English learning achievement. Our structural equation model further demonstrated that among all the predictors included in the EVT framework, the expectancy component had the largest total effect on language learning outcomes. This finding converges to the proposition suggested in this framework that expectancy has a stronger prediction power than value for the achievement aspect of learning (Schunk et al., 2008; Wigfield & Eccles, 2000). However, the role of the value component in language learning is less clear, although it was found to be significantly influenced by the participants' perception of technology use for language learning. Proponents of the EVT framework (Eccles & Wigfield, 1995, 2002; Wigfield & Eccles, 2000) have suggested that this component could predict learners' future engagement in the target task, which may be an important aspect of language learning in the long term; however, further empirical evidence is needed to prove this proposition in the context of language learning.

It is noteworthy that the present study is one of the first to adopt the EVT framework and investigate the role of its two main components (expectancy and value), along with other learner-related variables in language learning achievement. Our model, as presented in the *Results* section, had acceptable model fit indices, and the two components were correlated with other variables to varying degrees. To summarize, the results of our structural model suggest that the EVT framework can be applied to account for an achievement in the area of language learning, which can be considered as an important theoretical contribution.

Pedagogical and language educational policy implications

Based on the aforementioned findings of the current study, in this section, we provide pedagogical and language educational policy implications. First, regarding learners' SES background, as this variable was found to influence language learning achievement via mediators such as the perception of technology use and motivational factors, albeit weakly, it appears that language learners with a lower SES background and less access to technologies are rather disadvantaged (for a similar finding, see Kormos & Kiddle, 2013). In this regard, we largely concur with Ghorbani and Golparvar (2020), who suggested that 'these groups of learners are in more desperate need of fostering their motivational and efficacy beliefs to enhance their readiness for autonomous language learning with technology' (p. 622). To this end, creating in-class conditions optimized for CALL as well as after-school seminars on technology resources directed toward fostering learners' self-efficacy in using technologies may help alleviate such learners' disadvantages originating from their low SES background. We believe that such efforts must be driven by the macro-level educational policy of larger educational organizations and personnel, including government officials and policy makers, with a clear goal of mitigating the negative effects of the digital divide on language learning.

Second, regarding motivation, we suggest that teachers and parents should be made aware of the importance of raising learners' expectancy that they will be successful in language learning. This may be of considerable importance for those students who have lower beliefs about their competence to tackle target language learning tasks and are therefore in need of encouragement from people in their immediate learning environment.

Third, regarding learners' perception of technology use in CALL, teachers and educators could be more informed of the importance of learners' perception of technology use for language learning and its association with motivation and language learning achievement, as revealed in this study. Together with the previous finding on the important role that teachers play in language learners' perception and adoption of technology resources (Lai, 2015), this also implies that teachers and educators should also improve their own knowledge of technology use in language learning and be mindful that learners' perception of technology use does affect

their successful language learning. In this regard, teachers and parents should make an effort to create computer-assisted learning environments in class and at home, aim to enhance learners' self-efficacy of technology use, and help them recognize the usefulness of technologies in language learning.

Limitations and future directions

We now turn to the limitations of the present study, along with suggestions for future research. First, our participants were Korean college students learning English as a foreign language; they had been learning English for several years at school and in college before their participation in the present study and had already had good proficiency (average of B2 on CEFR). This suggests they were likely to have some notion of their own success in English learning. Therefore, our proposed SEM model and the identified relationships among the target variables may not be applicable to other language learners in different contexts. For example, in our model, mother's education level was not significantly correlated with the dependent variables, such as the perception of technology use for language learning and the two components of motivation (expectancy and value). However, mother's education level may have stronger predictive power in other language learning contexts, as found, for example, in the Chinese contexts by Butler (2014). Second, the present study did not investigate the role of other cognition-related variables in the complex relationships among the variables included. For example, variables such as language learning strategies, working memory, and learning styles may influence the structural paths and relationships examined in this study. We expect that future research with a larger sample size and a wider range of variables, including those listed above, will paint a clearer picture of the relationships among various factors, such as cognition, motivation, and achievement. Finally, the present study was an observational study by nature, implying that we cannot claim causality in the relationships among the target variables. Future studies may adopt a randomized control trial approach to assign participants to either computer-assisted or traditional instructional environments and may examine how this difference influences their motivation and language learning achievement.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix: Questionnaire items on the perception of technology use and motivational factors

< Perception of technology use – adapted from Lai & Gu, 2011; Lai et al., 2012 >

* Facilitation conditions (3 items)

1. I have the resources necessary to use technologies for English learning.
2. I have the knowledge necessary to use technologies for English learning.
3. When I need help using technology to enhance English learning, someone is there to help me.

* Technology self-efficacy (3 items)

4. I am confident in my ability to use technologies effectively for English learning.
5. I am confident in my ability to select appropriate technologies for my English learning needs.

6. I am confident in my ability to use technologies to create enjoyable English learning experiences.

* Perceived usefulness (4 items)

7. I think that using technologies for English learning improves my learning experience and environment.
8. I think that using technologies for English learning helps monitor my learning progress.
9. I think that using technologies for English learning sustains or enhances my motivation and interest in learning English.
10. I think that using technologies for English learning expands my language use opportunities.

< Motivational factors – adapted from Eccles and Wigfield (1995) >

* Task value (7 items)

11. In general, I find working on English assignments (very boring, very interesting).
12. How much do you like learning English (not very much, very much)?
13. Is the amount of effort it will take to do well in English worthwhile to you (not very worthwhile, very worthwhile)?
14. To me, being good at English is (not at all important, very important).
15. How important is it to you to get good results in English (not at all important, very important)?
16. How useful is learning English for what you want to do after you graduate and go to work (not very useful, very useful)?
17. How useful is what you learn in English for your daily life outside school (not at all useful, very useful)?

* Expectancy (5 items)

18. Compared to other students, how well do you expect to do in English this year (much worse than other students, much better than other students)?
19. How well do you think you will do in your English this year (very poorly, very well)?
20. How good at English are you (not at all good, very good)?
21. If you were to order all the students in your class from the worst to the best in English, where would you put yourself (the worst, the best)?
22. How have you been doing in English this year (very poorly, very well)?